Data-Backed Journal-Style Replication of *Something Old, Something New*

Replication Paper Project

2026-04-04

Abstract

This paper presents a local calculated replication of Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 2,174 observations across 162 entities, and the batch headline estimate is 0.0052 for search_scope with p = 0.6153. The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: search depth; search scope; new product introduction; firm-year innovation; local analogue

1 Introduction

This replication begins from a paper-first reading of Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether search depth and search scope are distinct dimensions and how

they shape subsequent innovation. The study is worth revisiting because The replication matters because the original paper's distinction between search depth and search scope is widely cited, yet the empirical mechanism depends heavily on how those constructs are operationalized. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted. The executable evidence base used in this manuscript contains 2,174 observations across 162 entities and spans 1986 to 2015 in the local executable panel.. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets. The paper's contribution is to recover the two-dimensional search logic in an executable patent panel and show where the patent-based analogue departs from the original product-outcome result. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions.

2 Original Study, Theory, and Replication Boundary

The focal source article is [1]. The original study can be summarized as a mechanism paper with a sharp empirical core. The original paper distinguishes search depth from search scope. The study is longitudinal and firm-year based. Innovation quantity and quality are both part of the argument. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data. In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue. The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local

replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design.

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Depth and scope are distinct dimensions. The local reconstruction therefore adopts the following bounded decision: The local panel maps depth and scope to executable recombination constructs. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The original design is firm-year based. The local reconstruction therefore adopts the following bounded decision: The firm-year structure is preserved exactly. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Innovation outcomes include quantity and quality. The local reconstruction therefore adopts the following bounded decision: Patent counts and patent-impact totals substitute for product introductions. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline.

DirectSourceFact	LocalReplicationDecision
Depth and scope are distinct dimensions.	The local panel maps depth and scope to executable recombination constructs.
The original design is firm-year based.	The firm-year structure is preserved exactly.
Innovation outcomes include quantity and quality.	Patent counts and patent-impact totals substitute for product introductions.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Firm-year panel aggregated from the local patent-level recombination dataset. Search depth measured from prior organizational recombination experience. Search scope measured from breadth of recombination in the current year. Operationally, the reconstruction moves through the following file chain: Analysis panel is backed by the local file katila_ahuja_2002_analysis_dataset.csv; Coefficient export is backed by the local file katila_ahuja_2002_model_coefficients.csv; Sample summary is backed by the local file katila_ahuja_2002_sample_summary.csv. The resulting executable panel supplies 2,174 observations across 162 entities and covers 1986 to 2015 in the local executable panel.. Every numeric claim in the results section is intended to trace back to one of those verified files rather than to a hand-copied number from the article. This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending

that the original archive has been fully recreated. The local analogue aggregates the patent-level recombination panel to firm-years and uses patent-based outcomes in place of the source paper's product-introduction outcomes. The local sample frame is Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions. The observed time window in the executable panel is 1986 to 2015 in the local executable panel. The outcome is Next-year patent quantity and next-year total patent impact. The main executable panel used in the paper is katila_ahuja_2002_analysis_dataset.csv and the principal coefficient export is katila_ahuja_2002_model_coefficients.csv .

File	Role
katila_ahuja_2002_analysis_dataset.csv	Firm-year search-behavior panel
katila_ahuja_2002_model_coefficients.csv	Quantity and impact coefficient table
katila_ahuja_2002_sample_summary.csv	Sample statistics

Core local datasets and result files used in the current paper.

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Katila and Ahuja (2002). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument. The analysis panel used in this paper is katila_ahuja_2002_analysis_dataset.csv. Its role is to estimate the local model and regenerate descriptive statistics. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local ma-

chine details. The coefficient export used in this paper is katila_ahuja_2002_model_coefficients.csv. Its role is to recover the reported local coefficients and p-values. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. The sample summary used in this paper is katila_ahuja_2002_sample_summary.csv. Its role is to audit sample construction and raw grouped moments. This file is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. By naming the dataset rather than printing a filesystem path, the paper keeps the emphasis on evidence and function rather than on local machine details. Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim.

Stage	NeededFor	Dataset	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	KatilaAhujaamj02.txt	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	katila_ahuja_2002_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	katila_ahuja_2002_model_coefficients.csv	Yes	Executable local result file

Stage	NeededFor	Dataset	Exists	Status
Sample summary	Audit sample construction and raw grouped moments	katila_ahuja_2002_sample_summary.csv	Yes	Executable local result file

Verified current files used to support the paper's calculations.

3.2 Sample Reconstruction

The local analysis file contains 2,174 observations drawn from 162 entities. The observed time span runs from 1,986 to 2,015. Selected sample moments are search_depth = 0.6637, search_scope = 4.2347, next_patents = 10.6026 . These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed.

Statistic	Value
rows	2174.0000
unique_firms	162.0000
year_min	1986.0000
year_max	2015.0000
mean_search_depth	0.6637
mean_search_scope	4.2347
mean_next_patents	10.6026

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Search depth is operationalized locally as Mean prior organizational experience and appears in the executable dataset as search_depth. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Average the current-year patent-level organizational experience within GVKEY-year.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Search scope is operationalized locally as Mean breadth and appears in the executable dataset as search_scope. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Average current-year breadth within GVKEY-year.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Current innovation scale is operationalized locally as Current-year patent count and appears in the executable dataset as current_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Count current-year patents.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point

where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Next-period quantity is operationalized locally as Next-year patent count and appears in the executable dataset as next_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Shift the patent count one year forward within GVKEY.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity. The construct Next-period impact is operationalized locally as Next-year total impact and appears in the executable dataset as next_total_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Shift the impact total one year forward within GVKEY.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Search depth	Mean prior organizational experience	search_depth	Average the current-year patent-level organizational experience within GVKEY-year.
Search scope	Mean breadth	search_scope	Average current-year breadth within GVKEY-year.
Current innovation scale	Current-year patent count	current_patents	Count current-year patents.
Next-period quantity	Next-year patent count	next_patents	Shift the patent count one year forward within GVKEY.
Next-period impact	Next-year total impact	next_total_impact	Shift the impact total one year forward within GVKEY.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Katila and Ahuja (2002). The point is not to rehearse notation for its own sake, but to show how

the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Search depth, the manuscript uses Mean prior organizational experience under the exact local implementation search_depth. The derivation step is: Average the current-year patent-level organizational experience within GVKEY-year.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Search scope, the manuscript uses Mean breadth under the exact local implementation search_scope. The derivation step is: Average current-year breadth within GVKEY-year.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Current innovation scale, the manuscript uses Current-year patent count under the exact local implementation current_patents. The derivation step is: Count current-year patents.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Next-period quantity, the manuscript uses Next-year patent count under the exact local implementation next_patents. The derivation step is: Shift the patent count one year forward within GVKEY.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Next-period impact, the manuscript uses Next-year total impact under the exact local implementation next_total_impact. The derivation step is: Shift the impact total one year forward within GVKEY.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Search depth has mean 0.6637, standard deviation 0.6622, and an interquartile range from 0.1386 to 0.9635; Search scope has mean 4.2347, standard deviation 2.8896, and an interquartile range from 2.8182 to 4.6667; Current patents has mean 10.5437, standard deviation 20.1122, and an interquartile range from 2.0000 to 11.0000; Current total impact has mean 56.5745, standard deviation 142.3809, and an interquartile range from 5.0000 to 47.0000 . The time-series view is equally informative. V1 changes from 100.0000 in the first observed period to 35.0000 in the last

observed period; V2 changes from 0.2522 in the first observed period to 1.2033 in the last observed period; V3 changes from 2.7740 in the first observed period to 6.8650 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output. The regressor-quantile analysis shows that mean outcomes range from 5.3149 to 17.4105 across the support of the focal regressor, while the mean regressor itself ranges from 0.1128 to 1.6987. That pattern is useful because it shows what the formal model is trying to account for in the current dataset. The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is.

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Search depth has missing share 0.00; Search scope has missing share 0.00; Current patents has missing share 0.00 . At the other end of the support, Search depth remains close to complete with missing share 0.00; Search scope remains close to complete with missing share 0.00; Current patents remains close to complete with missing share 0.00 . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively. Coverage across entities is also visible in the current panel rather than assumed. 10499 contributes 30 observations; 10983 contributes 30 observations; 1300 contributes 30 observations; 2285 contributes 30 observations; 2403 contributes 30 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities. The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 0.1128 and the mean outcome is 5.3149 across 870 observations. In the highest observed quantile, the corresponding values are 1.6987, 14.1885, and 435 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells. These

data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because the original paper's distinction between search depth and search scope is widely cited, yet the empirical mechanism depends heavily on how those constructs are operationalized. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication.

Variable	Missing	MissingShare
Search depth	0.0000	0.0000
Search scope	0.0000	0.0000
Current patents	0.0000	0.0000
Current total impact	0.0000	0.0000
Mean field experience	0.0000	0.0000
Next-year patents	0.0000	0.0000
Next-year total impact	0.0000	0.0000

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Search depth	2174.0000	0.6637	0.6622	0.1386	0.5493	0.9635	0.0000	4.76
Search scope	2174.0000	4.2347	2.8896	2.8182	3.4264	4.6667	2.0000	52.00
Current patents	2174.0000	10.5437	20.1122	2.0000	5.0000	11.0000	1.0000	399.00
Current total impact	2174.0000	56.5745	142.3809	5.0000	17.0000	47.0000	0.0000	3058.00
Mean field experience	2174.0000	1.4705	1.1482	0.6931	1.3041	1.9634	0.0000	7.55
Next-year patents	2174.0000	10.6026	21.2666	2.0000	4.0000	11.0000	1.0000	399.00
Next-year total impact	2174.0000	56.6297	144.9651	5.0000	16.0000	46.0000	0.0000	3058.00

Main descriptive statistics discussed in the body of the paper.

Period	V1	V2	V3	V4	V5
1986.0000	100.0000	0.2522	2.7740	7.7000	33.2900
1987.0000	103.0000	0.3220	2.9110	10.2136	45.3592
1988.0000	105.0000	0.3484	3.0106	13.1714	51.6571
1989.0000	98.0000	0.4390	3.1399	13.5102	61.0510
1990.0000	111.0000	0.4669	3.0152	11.9640	56.5225
1991.0000	109.0000	0.5001	3.1953	12.2110	61.1927
1992.0000	98.0000	0.4820	3.3594	13.5408	77.0510
1993.0000	100.0000	0.5504	3.4467	13.2200	82.5600
1994.0000	104.0000	0.5512	3.6508	11.6442	72.5481
1995.0000	98.0000	0.5892	3.7678	6.9490	45.8367
1996.0000	83.0000	0.6681	4.1041	7.6265	45.4217
1997.0000	80.0000	0.8318	4.0021	4.8500	30.2750
1998.0000	65.0000	0.6553	3.9114	9.2000	50.7077
1999.0000	78.0000	0.6504	3.7812	8.1410	45.8590
2000.0000	73.0000	0.8168	3.9949	8.3014	42.4384
2001.0000	72.0000	0.7542	4.2585	8.8056	42.2083
2002.0000	65.0000	0.8268	4.4429	8.0462	36.2308
2003.0000	62.0000	0.8369	4.8452	5.2581	19.5000
2004.0000	59.0000	0.7436	4.8832	5.0847	23.0678
2005.0000	48.0000	0.7782	5.1188	6.8125	33.2083
2006.0000	51.0000	0.8915	5.2182	4.9216	17.0392
2007.0000	43.0000	0.8791	6.2118	12.6047	48.6047
2008.0000	38.0000	0.9390	5.6773	5.2895	25.4737
2009.0000	43.0000	1.0399	6.1656	11.3488	49.2093
2010.0000	50.0000	0.9987	6.7648	12.9600	83.4800
2011.0000	51.0000	0.9412	7.1105	20.7059	83.2157
2012.0000	52.0000	1.0293	6.9358	16.5192	88.8462

Period	V1	V2	V3	V4	V5
2013.0000	50.0000	1.0361	6.7054	14.3400	180.3600
2014.0000	50.0000	1.1223	6.3393	21.3600	141.9400
2015.0000	35.0000	1.2033	6.8650	14.8857	57.1429

Observed time patterns in the current dataset.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	870.0000	0.1128	5.3149
2.0000	506.0000	0.5658	11.7273
3.0000	363.0000	0.8803	17.4105
4.0000	435.0000	1.6987	14.1885

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Search depth	Search scope	Current patents	Current total impact	Mean field experience	Next-year patents	Next-year total impact
Search depth	1.000	-0.011	0.184	0.138	0.735	0.158	0.106
Search scope	-0.011	1.000	0.010	0.146	-0.242	0.004	0.092
Current patents	0.184	0.010	1.000	0.612	0.062	0.746	0.513
Current total impact	0.138	0.146	0.612	1.000	0.042	0.482	0.670
Mean field experience	0.735	-0.242	0.062	0.042	1.000	0.046	0.020
Next-year patents	0.158	0.004	0.746	0.482	0.046	1.000	0.630

Compact correlation structure for the main variables.

Entity	Observations	FirstPeriod	LastPeriod
10499.0000	30.0000	1986.0000	2015.0000
10983.0000	30.0000	1986.0000	2015.0000
1300.0000	30.0000	1986.0000	2015.0000
2285.0000	30.0000	1986.0000	2015.0000
2403.0000	30.0000	1986.0000	2015.0000
3532.0000	30.0000	1986.0000	2015.0000
4087.0000	30.0000	1986.0000	2015.0000
4321.0000	30.0000	1986.0000	2015.0000
4503.0000	30.0000	1986.0000	2015.0000
4839.0000	30.0000	1986.0000	2015.0000
5047.0000	30.0000	1986.0000	2015.0000
5439.0000	30.0000	1986.0000	2015.0000
6066.0000	30.0000	1986.0000	2015.0000
6266.0000	30.0000	1986.0000	2015.0000
7435.0000	30.0000	1986.0000	2015.0000
8762.0000	30.0000	1986.0000	2015.0000
2817.0000	29.0000	1986.0000	2015.0000
3835.0000	29.0000	1986.0000	2015.0000
6435.0000	29.0000	1986.0000	2015.0000
7257.0000	29.0000	1986.0000	2015.0000

Most observed entities in the current dataset.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is Firm and year fixed-effects models for next-year patent counts and next-year total impact using local depth, scope, and their interaction. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that

embody the source paper's central mechanism. Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest. The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions., and the empirical window is 1986 to 2015 in the local executable panel.. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample. The second set of decisions concerns measurement and specification. The outcome is Next-year patent quantity and next-year total patent impact., and the main explanatory design is Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question. The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Current innovation scale and field-experience controls with firm and year effects.. The crucial exclusion or substitution rule is: The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability. A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether search depth and search scope are distinct dimensions and how they shape subsequent innovation.
Unit of analysis	Firm-year
Sample frame	Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions.
Time window	1986 to 2015 in the local executable panel.
Outcome definition	Next-year patent quantity and next-year total patent impact.
Main explanatory design	Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.
Controls and fixed effects	Current innovation scale and field-experience controls with firm and year effects.

DecisionPoint	LocalSpecification
Exclusion or substitution logic	The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Firm-year panel aggregated from the local patent-level recombination dataset for firms with usable year-to-year transitions. over 1986 to 2015 in the local executable panel.. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Next-year patent quantity and next-year total patent impact. and the main explanatory design is Firm and year fixed-effects models relating next-period innovation to current search depth, scope, and their interaction.. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Current innovation scale and field-experience controls with firm and year effects. The main exclusion or substitution rule is The local executable design substitutes patent-based future outcomes for product-introduction outcomes and keeps only firm-years with usable lag-lead construction. This is the point at which the paper most clearly separates direct

source facts from local choices, and that separation is intentional rather than apologetic.

$$E[\text{NextPatents}_{it} \mid X_{it}] = \exp\left(\beta_1 \text{Depth}_{it} + \beta_2 \text{Depth}_{it}^2 + \beta_3 \text{Scope}_{it} + \beta_4 \text{Depth}_{it} \times \text{Scope}_{it} + \alpha_i + \gamma_t\right), \tag{1}$$

$$\log(1 + \text{NextImpact}_{it}) = \theta_1 \text{Depth}_{it} + \theta_2 \text{Depth}_{it}^2 + \theta_3 \text{Scope}_{it} + \theta_4 \text{Depth}_{it} \times \text{Scope}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{2}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.0052 on search_scope (p = 0.6153). The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. Log(1 + current patents) is estimated at 0.6357 with p = ¡0.001. Depth x scope is estimated at -0.0216 with p = 0.0982. Search scope is estimated at 0.0184 with p = 0.1671. The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

model	Term	estimate	std_error	p_value
katila_ahuja_next_impact	Search depth	-0.1070	0.1214	0.3854
katila_ahuja_next_impact	Search depth squared	0.0388	0.0322	0.2375
katila_ahuja_next_impact	Search scope	0.0184	0.0130	0.1671
katila_ahuja_next_impact	Log(1 + current patents)	0.6357	0.0617	0.0000
katila_ahuja_next_impact	Mean field experience	0.0271	0.0350	0.4444
katila_ahuja_next_impact	Depth x scope	-0.0216	0.0126	0.0982

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log(1 + current patents) is estimated at 0.6357 with a p-value of ¡0.001; Depth x scope is estimated at -0.0216 with a p-value of 0.0982; Search scope is estimated at 0.0184 with a p-value of 0.1671; Search depth squared is estimated at 0.0388 with a p-value of 0.2375 . Within the focal specification, the sharpest local signal comes from Log(1 + current patents), which is estimated at 0.6357 with p = ¡0.001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment. The scale of the underlying variables also matters for interpretation. The outcome Next-year patents has mean 10.6026 and standard deviation 21.2666 in the local panel, while the main regressor Search depth has mean 0.6637 and standard deviation 0.6622. A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 5.3149 to 17.4105 while the mean regressor ranges from 0.1128 to 1.6987. This does not substitute for the model, but it shows how much raw variation the specification must explain. Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local quantity model points to a nonlinear depth effect and a weaker role for search scope than the source paper emphasizes. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation.

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification. Within katila_ahuja_next_impact, the leading coefficient family reads as follows: Log(1 + current patents) = 0.6357 (p = ¡0.001); Depth x scope = -0.0216 (p = 0.0982); Search scope = 0.0184 (p = 0.1671); Search depth squared = 0.0388 (p = 0.2375); Search depth = -0.1070 (p = 0.3854). The largest positive estimate in this specification is Log(1 + current patents) at 0.6357, while the largest negative estimate is Search depth at -0.1070. Reading the specification this way matters because

it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. Within katila_ahuja_next_patents, the leading coefficient family reads as follows: Log(1 + current patents) = 0.6654 (p = ¡0.001); Search depth squared = 0.1059 (p = ¡0.001); Search depth = -0.2243 (p = 0.0250); Mean field experience = -0.0429 (p = 0.1018); Depth x scope = -0.0115 (p = 0.2651). The largest positive estimate in this specification is Log(1 + current patents) at 0.6654, while the largest negative estimate is Search depth at -0.2243. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure. This fuller coefficient reading is especially important for Katila and Ahuja (2002) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text.

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In katila_ahuja_next_patents, the most informative focal term is Log(1 + current patents) with estimate 0.6654 and p = ¡0.001; In katila_ahuja_next_impact, the most informative focal term is Log(1 + current patents) with estimate 0.6357 and p = ¡0.001 . The katila_ahuja_next_patents specification contributes its own reading to the replication. Log(1 + current patents) is estimated at 0.6654 with p = ¡0.001 and is classified as a strong signal; Search depth squared is estimated at 0.1059 with p = ¡0.001 and is classified as a strong signal; Search depth is estimated at -0.2243 with p = 0.0250 and is classified as a moderate signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. The katila_ahuja_next_impact specification contributes its own reading to the replication. Log(1 + current patents) is estimated at 0.6357 with p = ¡0.001 and is classified as a strong signal; Depth x scope is estimated at -0.0216 with p = 0.0982 and is classified as a suggestive signal; Search scope is estimated at 0.0184 with p = 0.1671 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more

clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient. This model comparison matters for Katila and Ahuja (2002) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition.

Term	Model	Estimate	PValue	Signal	Reading
Search depth	katila_ahuja_next_patents	-0.2243	0.0250	Moderate	Negative estimate with p = 0.0250.
Search depth squared	katila_ahuja_next_patents	0.1059	0.0004	Strong	Positive estimate with p = ¡0.001.
Search scope	katila_ahuja_next_patents	0.0052	0.6153	Weak	Positive estimate with p = 0.6153.
Log(1 + current patents)	katila_ahuja_next_patents	0.6654	0.0000	Strong	Positive estimate with p = ¡0.001.
Mean field experience	katila_ahuja_next_patents	-0.0429	0.1018	Weak	Negative estimate with p = 0.1018.

Term	Model	Estimate	PValue	Signal	Reading
Depth x scope	katila_ahuja_next_patents	-0.0115	0.2651	Weak	Negative estimate with p = 0.2651.
Search depth	katila_ahuja_next_impact	-0.1070	0.3854	Weak	Negative estimate with p = 0.3854.
Search depth squared	katila_ahuja_next_impact	0.0388	0.2375	Weak	Positive estimate with p = 0.2375.
Search scope	katila_ahuja_next_impact	0.0184	0.1671	Weak	Positive estimate with p = 0.1671.
Log(1 + current patents)	katila_ahuja_next_impact	0.6357	0.0000	Strong	Positive estimate with p = ¡0.001.
Mean field experience	katila_ahuja_next_impact	0.0271	0.4444	Weak	Positive estimate with p = 0.4444.
Depth x scope	katila_ahuja_next_impact	-0.0216	0.0982	Suggestive	Negative estimate with p = 0.0982.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

A separate model-fit export is not available for this paper, so the interpretation relies on the coefficient file and the observed distribution of the current dataset.

term	estimate	std_error	statistic	p_value	model
search_depth	-0.2243	0.1001	-2.2411	0.0250	katila_ahuja_next_pate
I(I(search_depth^2))	0.1059	0.0299	3.5451	0.0004	katila_ahuja_next_pate
search_scope	0.0052	0.0104	0.5025	0.6153	katila_ahuja_next_pate
log1p(current_patents)	0.6654	0.0652	10.2026	0.0000	katila_ahuja_next_pate
mean_field_exp	-0.0429	0.0262	-1.6360	0.1018	katila_ahuja_next_pate
search_depth:search_scope	-0.0115	0.0103	-1.1143	0.2651	katila_ahuja_next_pate
search_depth	-0.1070	0.1214	-0.8812	0.3854	katila_ahuja_next_impa
I(I(search_depth^2))	0.0388	0.0322	1.2061	0.2375	katila_ahuja_next_impa
search_scope	0.0184	0.0130	1.4170	0.1671	katila_ahuja_next_impa
log1p(current_patents)	0.6357	0.0617	10.3089	0.0000	katila_ahuja_next_impa
mean_field_exp	0.0271	0.0350	0.7753	0.4444	katila_ahuja_next_impa
search_depth:search_scope	-0.0216	0.0126	-1.7086	0.0982	katila_ahuja_next_impa

Full coefficient table from the current local estimation output.

7 Comparison to the Published Study

The comparison to the published paper is intentionally narrow. The local analogue preserves the paper's dual-dimension logic but does not produce the same relative strength for depth and scope. The quantity model highlights a nonlinear depth pattern, while search scope is comparatively muted in the executable patent-based panel. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about

exact numerical replication. This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation.

8 Discussion

The local analogue preserves the paper's dual-dimension logic but does not produce the same relative strength for depth and scope. The quantity model highlights a nonlinear depth pattern, while search scope is comparatively muted in the executable patent-based panel. Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly. The first limitation concerns measurement. The local analogue uses patent outcomes rather than product-introduction outcomes. The second limitation concerns sample translation. Search variables are mapped from recombination measures already in the workspace. The third limitation concerns interpretation. The exercise is strongest on conceptual mechanism and panel structure, not exact outcome identity.

9 Conclusion

The paper's contribution is to recover the two-dimensional search logic in an executable patent panel and show where the patent-based analogue departs from the original product-outcome result. The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available.

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value
rows	2174.0000
unique_firms	162.0000
year_min	1986.0000
year_max	2015.0000
mean_search_depth	0.6637
mean_search_scope	4.2347
mean_next_patents	10.6026

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Search depth	2174.0000	0.6637	0.6622	0.1386	0.5493	0.9635	0.0000	4.76
Search scope	2174.0000	4.2347	2.8896	2.8182	3.4264	4.6667	2.0000	52.00
Current patents	2174.0000	10.5437	20.1122	2.0000	5.0000	11.0000	1.0000	399.00
Current total impact	2174.0000	56.5745	142.3809	5.0000	17.0000	47.0000	0.0000	3058.00
Mean field experience	2174.0000	1.4705	1.1482	0.6931	1.3041	1.9634	0.0000	7.55
Next-year patents	2174.0000	10.6026	21.2666	2.0000	4.0000	11.0000	1.0000	399.00
Next-year total impact	2174.0000	56.6297	144.9651	5.0000	16.0000	46.0000	0.0000	3058.00

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Search depth	Search scope	Current patents	Current total impact	Mean field experience	Next-year patents	Next-year total impact
Search depth	1.000	-0.011	0.184	0.138	0.735	0.158	0.106
Search scope	-0.011	1.000	0.010	0.146	-0.242	0.004	0.092
Current patents	0.184	0.010	1.000	0.612	0.062	0.746	0.513
Current total impact	0.138	0.146	0.612	1.000	0.042	0.482	0.670
Mean field experience	0.735	-0.242	0.062	0.042	1.000	0.046	0.020
Next-year patents	0.158	0.004	0.746	0.482	0.046	1.000	0.630
Next-year total impact	0.106	0.092	0.513	0.670	0.020	0.630	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	V1	V2	V3	V4	V5
1986.0000	100.0000	0.2522	2.7740	7.7000	33.2900
1987.0000	103.0000	0.3220	2.9110	10.2136	45.3592
1988.0000	105.0000	0.3484	3.0106	13.1714	51.6571
1989.0000	98.0000	0.4390	3.1399	13.5102	61.0510
1990.0000	111.0000	0.4669	3.0152	11.9640	56.5225
1991.0000	109.0000	0.5001	3.1953	12.2110	61.1927
1992.0000	98.0000	0.4820	3.3594	13.5408	77.0510
1993.0000	100.0000	0.5504	3.4467	13.2200	82.5600
1994.0000	104.0000	0.5512	3.6508	11.6442	72.5481
1995.0000	98.0000	0.5892	3.7678	6.9490	45.8367
1996.0000	83.0000	0.6681	4.1041	7.6265	45.4217
1997.0000	80.0000	0.8318	4.0021	4.8500	30.2750
1998.0000	65.0000	0.6553	3.9114	9.2000	50.7077
1999.0000	78.0000	0.6504	3.7812	8.1410	45.8590
2000.0000	73.0000	0.8168	3.9949	8.3014	42.4384
2001.0000	72.0000	0.7542	4.2585	8.8056	42.2083
2002.0000	65.0000	0.8268	4.4429	8.0462	36.2308
2003.0000	62.0000	0.8369	4.8452	5.2581	19.5000
2004.0000	59.0000	0.7436	4.8832	5.0847	23.0678
2005.0000	48.0000	0.7782	5.1188	6.8125	33.2083
2006.0000	51.0000	0.8915	5.2182	4.9216	17.0392
2007.0000	43.0000	0.8791	6.2118	12.6047	48.6047
2008.0000	38.0000	0.9390	5.6773	5.2895	25.4737
2009.0000	43.0000	1.0399	6.1656	11.3488	49.2093

Period	V1	V2	V3	V4	V5
2010.0000	50.0000	0.9987	6.7648	12.9600	83.4800
2011.0000	51.0000	0.9412	7.1105	20.7059	83.2157
2012.0000	52.0000	1.0293	6.9358	16.5192	88.8462
2013.0000	50.0000	1.0361	6.7054	14.3400	180.3600
2014.0000	50.0000	1.1223	6.3393	21.3600	141.9400
2015.0000	35.0000	1.2033	6.8650	14.8857	57.1429

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	870.0000	0.1128	5.3149
2.0000	506.0000	0.5658	11.7273
3.0000	363.0000	0.8803	17.4105
4.0000	435.0000	1.6987	14.1885

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
search_depth	-0.2243	0.1001	-2.2411	0.0250	katila_ahuja_next_pate
I(I(search_depth^2))	0.1059	0.0299	3.5451	0.0004	katila_ahuja_next_pate
search_scope	0.0052	0.0104	0.5025	0.6153	katila_ahuja_next_pate
log1p(current_patents)	0.6654	0.0652	10.2026	0.0000	katila_ahuja_next_pate
mean_field_exp	-0.0429	0.0262	-1.6360	0.1018	katila_ahuja_next_pate
search_depth:search_scope	-0.0115	0.0103	-1.1143	0.2651	katila_ahuja_next_pate
search_depth	-0.1070	0.1214	-0.8812	0.3854	katila_ahuja_next_impa
I(I(search_depth^2))	0.0388	0.0322	1.2061	0.2375	katila_ahuja_next_impa
search_scope	0.0184	0.0130	1.4170	0.1671	katila_ahuja_next_impa
log1p(current_patents)	0.6357	0.0617	10.3089	0.0000	katila_ahuja_next_impa
mean_field_exp	0.0271	0.0350	0.7753	0.4444	katila_ahuja_next_impa
search_depth:search_scope	-0.0216	0.0126	-1.7086	0.0982	katila_ahuja_next_impa

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Search depth	0.0000	0.0000
Search scope	0.0000	0.0000
Current patents	0.0000	0.0000
Current total impact	0.0000	0.0000
Mean field experience	0.0000	0.0000
Next-year patents	0.0000	0.0000
Next-year total impact	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

Entity	Observations	FirstPeriod	LastPeriod
10499.0000	30.0000	1986.0000	2015.0000
10983.0000	30.0000	1986.0000	2015.0000
1300.0000	30.0000	1986.0000	2015.0000
2285.0000	30.0000	1986.0000	2015.0000
2403.0000	30.0000	1986.0000	2015.0000
3532.0000	30.0000	1986.0000	2015.0000
4087.0000	30.0000	1986.0000	2015.0000
4321.0000	30.0000	1986.0000	2015.0000
4503.0000	30.0000	1986.0000	2015.0000
4839.0000	30.0000	1986.0000	2015.0000
5047.0000	30.0000	1986.0000	2015.0000
5439.0000	30.0000	1986.0000	2015.0000
6066.0000	30.0000	1986.0000	2015.0000
6266.0000	30.0000	1986.0000	2015.0000
7435.0000	30.0000	1986.0000	2015.0000

Entity	Observations	FirstPeriod	LastPeriod
8762.0000	30.0000	1986.0000	2015.0000
2817.0000	29.0000	1986.0000	2015.0000
3835.0000	29.0000	1986.0000	2015.0000
6435.0000	29.0000	1986.0000	2015.0000
7257.0000	29.0000	1986.0000	2015.0000

Top entities by local observation count.

References

[1] Riitta Katila and Gautam Ahuja. Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194, 2002.